Sharp subsidies for truthful production and network cooperation

Maxwell Flanagan — Claim and model comparison — 14 October 2026

Question for a specialist. Does the sharp joint coverage–incentive–budget theorem below follow from earlier work, and are its two proofs valid? This is an internal-review candidate, not a claim of external endorsement.

Fixed model. There are n fixed owners and d resources. Each owner reports $b_i \in \mathbb{R}^d_+$ to the same fixed ordered family $V = \{w^1, \dots, w^K\} \subseteq \mathbb{R}^d_+$:

$$f(c) = \min_{w \in V} w \cdot c, \qquad v_b(S) = f\left(\sum_{i \in S} b_i\right), \qquad T = v_b(N).$$

A single deterministic rule on every real report profile has nonnegative payments M_i, covers every coalition ($\sum_{i \in S} M_i \geq v_b(S)$), and is componentwise nondecreasing in each owner's entire bundle with others fixed. This means truthfulness for binding underreports and payment-only utility. There are no resource-use costs, overreports, coordinated-report, false-name, split/merge, other-owner or population guarantees. Owners are never replaced by strategic capacity units. A total-payment factor B corresponds to subsidy factor $B - 1$.

Proposed sharp theorem. For every integer $j \geq 2$ an eligible rule satisfies

$$\sum_i M_i \leq C_j(K) T, \qquad C_j(K) = j + \frac{\log K + 2 \log j + 1}{\log[j^3/(j^2 + j - 1)]}.$$

Choosing $j = \max\{2, \lceil \log K / [\log(2 + \log K)]^2 \rceil\}$ gives $(1 + o(1))T \log K / \log\log K$. Zero-value profiles receive zero. The worst-case optimum with *at most* K forms has the same leading constant one. The lower already holds for simple directed acyclic flow networks with at most K structural cuts and one strategic owner per edge. It is not an exact-count padding theorem for every integer K.

Upper proof to check. Delete whole owners independently with probability q; select a fixed-priority minimum *raw* form. Let p_i^q be the expected contribution capped at T conditional on deleting i, and m_i^q the expected on-minus-off production marginal. Both are own-monotone. The mixture $r_i^q = q p_i^q + (1 - q) m_i^q$ dominates a coalition-covering capped form allocation. For masked production g and total capped payment Z,

$$R = Z - g - (1 - q) g', \qquad \int_0^\eta R = \int_0^\eta Z + T - (1 - \eta) g(\eta) - 2 \int_0^\eta g.$$

A Chernoff union over the fixed forms bounds Z. The payment $M_i = j \int_0^{1/j} r_i^q \, dq$ is deterministic. Selection is uncapped; only payment contributions are capped.

Lower proof to check. Shared-fan modules keep the network simple and all auxiliary owners strategic. Actual coalition inequalities and multinomial weights $p_h \propto \lambda^h / h!$ transport only the changing owner's payment. With disjoint middle-edge and auxiliary budgets Y, Q,

$$A \leq (1 + \alpha)\mathbb{E}Y + (m/R)\mathbb{E}Q \leq \max\{1 + \alpha, m/R\} B \mathbb{E}T.$$

Taking $R = m$ preserves the constant. Cut counts and hierarchy parameters then give $\log K \sim h \log h$ and factor $(1 - o(1))h$.

Closest original-source comparisons.

Production and stability. Owen (1975) and Samet–Zemel (1984) establish dual-price/core ingredients. Moretti–Norde (2021) studies the weighted multi-glove game used as a special case. Fixed-game cost of stability is established. Our issue links different reports even though each individual production game has an efficient core.

Same disclosure model. Huang–Mei–Yang–Cao (2025 v1) uses binding capacity underreports and payment utility in flow games. Its efficient mechanisms and core-selection obstruction are the nearest strategic comparison. The present theorem keeps all coalition inequalities and allows subsidy; it does not inherit their extra incentive guarantees.

Different monotonicity. Immorlica–Mahdian–Mirrokni compares an unchanged owner's payment when other owners join. Brânzei–Solymosi–Tijs (2007) compares participating subgames of unit owners (final Definition 1, Theorems 5–6). Neither condition is silently identified with our same-owner bundle comparison.

Same own-resource condition, different technology. Amigo–Belzarena–Vaton (2016 author version) defines own-resource monotonicity and a strictly concave service model. The supplement's smooth strictly concave example fits that inspected model and refutes a universal extension of the reported projection simulations, not a proved theorem. The publisher final has not been compared in full. This diagnostic is not homogeneous and is separate from the main theorem.

Established tools versus proposed contribution. Supporting prices, max-flow/min-cut, marginal contributions, own-excluded selection, concentration, weighted counting, convex support theory and uniform convergence are established. The proposed contribution is the joint sharp all-profile guarantee and matching fixed-owner simple-network lower bound. The source search is bounded; historical priority remains unconfirmed.

What the result does not settle. K can be exponential in an encoded technology. The owner-count and resource-dimension gaps remain open. Exact enumeration and the supplementary certification procedure may require exponential preprocessing. The paper asserts neither a recognized major open-problem resolution nor a general theorem for private opportunity costs.

Review evidence and corrections. The main paper contains complete upper and lower proofs. The indexed supplement preserves the October research record and its supporting results. Prior scoped cross-model reviews and fresh independent Codex audits are documented by source hash; finite exact checks are corroboration, not formal proof verification. The 14 October preparation repairs an auxiliary two-owner example so its efficient alternative works on whole bundles, and makes $0 < \varepsilon < 1$ explicit in the lower bound's inversion. Neither changes the central theorem. Current cross-model quota exhaustion prevented a fresh Claude audit; no human review is claimed.

Requested assessment. Check the cap/endpoint/deletion identities; check symmetrization, auxiliary budgets, weighted transitions and cut inversion; identify any original theorem with the same fixed-owner, all-profile, full-coverage guarantee. Author affiliation, contact, declarations and final responsibility still need confirmation before any submission. No outreach or submission has been performed.